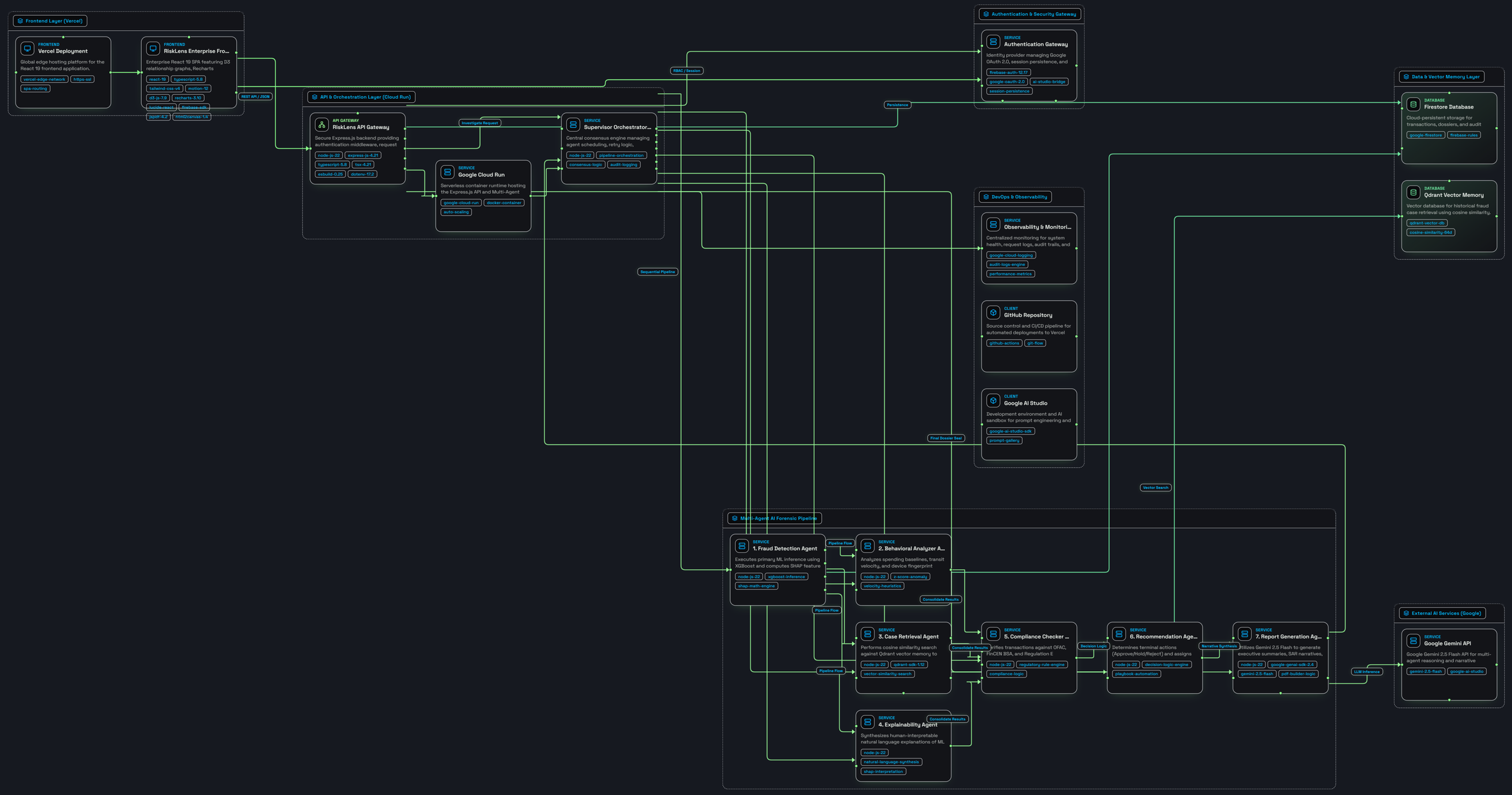

Frontend Layer (Vercel)

FRONTEND
Vercel Deployment
Global edge hosting platform for the React 19 frontend application.
vercel-edge-network · https-ssl · spa-routing

FRONTEND
RiskLens Enterprise Fro...
Enterprise React 19 SPA featuring D3 relationship graphs, Recharts
react-19 · typescript-5.8 · tailwind-css-v4 · motion-12 · d3-js-7.9 · recharts-3.10 · lucide-react · firebase-sdk · jspdf-4.2 · html2canvas-1.4

API & Orchestration Layer (Cloud Run)

API GATEWAY
RiskLens API Gateway
Secure Express.js backend providing authentication middleware, request
node-js-22 · express-js-4.21 · typescript-5.8 · tsx-4.21 · esbuild-0.25 · dotenv-17.2

SERVICE
Google Cloud Run
Serverless container runtime hosting the Express.js API and Multi-Agent
google-cloud-run · docker-container · auto-scaling

SERVICE
Supervisor Orchestrator...
Central consensus engine managing agent scheduling, retry logic,
node-js-22 · pipeline-orchestration · consensus-logic · audit-logging

Authentication & Security Gateway

SERVICE
Authentication Gateway
Identity provider managing Google OAuth 2.0, session persistence, and
firebase-auth-12.17 · google-oauth-2.0 · ai-studio-bridge · session-persistence

DevOps & Observability

SERVICE
Observability & Monitori...
Centralized monitoring for system health, request logs, audit trails, and
google-cloud-logging · audit-logs-engine · performance-metrics

CLIENT
GitHub Repository
Source control and CI/CD pipeline for automated deployments to Vercel
github-actions · git-flow

CLIENT
Google AI Studio
Development environment and AI sandbox for prompt engineering and
google-ai-studio-sdk · prompt-gallery

Data & Vector Memory Layer

DATABASE
Firestore Database
Cloud-persistent storage for transactions, dossiers, and audit
google-firestore · firebase-rules

DATABASE
Qdrant Vector Memory
Vector database for historical fraud case retrieval using cosine similarity.
qdrant-vector-db · cosine-similarity-64d

Multi-Agent AI Forensic Pipeline

SERVICE
1. Fraud Detection Agent
Executes primary ML inference using XGBoost and computes SHAP feature
node-js-22 · xgboost-inference · shap-math-engine

SERVICE
2. Behavioral Analyzer A...
Analyzes spending baselines, transit velocity, and device fingerprint
node-js-22 · z-score-anomaly · velocity-heuristics

SERVICE
3. Case Retrieval Agent
Performs cosine similarity search against Qdrant vector memory to
node-js-22 · qdrant-sdk-1.12 · vector-similarity-search

SERVICE
4. Explainability Agent
Synthesizes human-interpretable natural language explanations of ML
node-js-22 · natural-language-synthesis · shap-interpretation

SERVICE
5. Compliance Checker ...
...rifies transactions against OFAC, FinCEN BSA, and Regulation E
node-js-22 · regulatory-rule-engine · compliance-logic

SERVICE
6. Recommendation Age...
Determines terminal actions (Approve/Hold/Reject) and assigns
node-js-22 · decision-logic-engine · playbook-automation

SERVICE
7. Report Generation Ag...
...tilizes Gemini 2.5 Flash to generate executive summaries, SAR narratives,
node-js-22 · google-genai-sdk-2.4 · gemini-2.5-flash · pdf-builder-logic

External AI Services (Google)

SERVICE
Google Gemini API
Google Gemini 2.5 Flash API for multi-agent reasoning and narrative
gemini-2.5-flash · google-ai-studio